To Facilitate or not to Facilitate: Human and LLM Facilitator Tendencies in Online Discussions

Anonymous ACL submission

Abstract

Automating facilitation in online discussions is a long-standing social concern given the increasing time we spend on online spaces and the failure of content moderation approaches. While studies have been conducted on how to facilitate, none have answered the essential question of when to do so. A potential answer is using LLMs, which ostensibly make automated, large-scale intervention increasingly feasible. In this study, we examine when LLMs decide to facilitate by defining what facilitation is, observing when humans decide to facilitate, and comparing their decisions with those made by LLMs. To this end, we create PEFK, a corpus standardizing and aggregating all relevant facilitation datasets. We are the first to run a survey on facilitation timing, which we execute using expert facilitative participants and LLM-as-a-judge models. We discover that while humans are more cautious, LLMs are excessively eager to facilitate, although both are more certain when judging that facilitation is not needed. We then investigate whether this behavior can be corrected using alternative setups for LLMs and training ModernBert classifiers on established datasets, finding that the latter perform more reliably than the former, although current datasets result in relatively low performance.

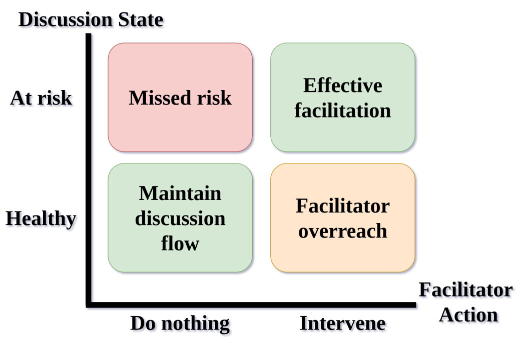

Figure 1: At any point in a discussion, the facilitator must decide whether to intervene or not. Not intervening when the situation calls for it may lead to the discussion worsening (e.g., topic derailment, escalation, missed participation). On the other hand, intervening when the discussion is already going well may lead to irritation among participants.

1 Introduction

Given the scale of social media networks and online spaces, facilitation needs to be automated to keep up with the amount of posted content (Small et al., 2023; Burton et al., 2024). So far, platforms have largely ignored the issue, preferring automated flag-and-remove mechanisms ("content moderation") (Korre et al., 2025), which are not sufficient (Cresci et al., 2022; Trujillo and Cresci, 2022). While recent work has undertaken the task of choosing *how* to facilitate (Chen et al., 2025; Schroeder et al., 2024; Park et al., 2012), there has been little interest in *when* to do so (Falk et al., 2021).

Deciding when to intervene in a discussion is a matter of great importance. Intervening too early or too frequently may lead to users expressing frustration at being constantly interrupted, while intervening too late may lead to a significant escalation and derailment of the discussion (Fig. 1, Korre et al. (2025); Gao et al. (2025)). Due to the sheer scale of modern online spaces automation is needed, which the field of Natural Language Processing (NLP) has attempted to implement using models to determine when an intervention is necessary. Encoder-Only (EncOn) classifiers were used to this end (Falk et al., 2021), although recently LLMs have been championed as the eventual solution to online facilitation (Korre et al., 2025; Small et al., 2023). However, previous work in LLM facilitation indicates that LLM facilitators are way too eager to intervene, rendering them unusable as autonomous facilitative agents (Tsirmpas et al., 2026). Yet, this finding has not been contrasted with human tendencies. In this study, we focus on the following Research Question (RQ):

RQ. *How and why do LLMs differ with humans when deciding when to facilitate?*

In order to answer the central research question we need to answer the following sub-questions:

What constitutes a facilitative intervention? There is currently no consensus on how to define a facilitative intervention. By examining relevant literature, we find three common definitions implicitly used in prior work; *escalation, professional facilitation*, and *facilitative comments* (§2.1).

Where do humans facilitate? While the majority of content where facilitation is needed is posted on online, written discussions, recent work predominantly focuses on *domain-specific oral discussion* scenarios (e.g., radio debates). Since datasets for discussion facilitation are generally few, small, and straddle radically different domains (e.g., debates, forums, deliberations), we gather all datasets relating to any of the three possible definitions of the term and compile the "Prosocial and Effective Facilitation in Konversations (PEFK)", a standardized dataset for discussion facilitation with a unified data and label schema (§3).

How do humans facilitate? We create and execute a survey task designed to gauge human facilitative patterns by sampling 1226 discussion chunks from our dataset, and asking 10 expert facilitative participants whether they would intervene as facilitators. We find that human participants tend to take a cautious approach to facilitation, generally preferring not to intervene and expressing greater confidence in not doing so (Finding 1). Additionally, human participants may pay attention to different spans of a discussion, which can lead them to choose different courses of action regarding facilitation.

How does LLM facilitation differ? We repeat the survey with six open-source LLMs. Unlike humans, LLMs tend to over-intervene, confirming previous studies (Tsirmpas et al., 2026) and rely more heavily on positive reinforcement. However, compared to humans, they may be more consistent between themselves, hinting that they are likely focusing on similar spans of the discussion. We also find that, much like humans, determining when not to facilitate is easy–the opposite is not (Findings 2,3).

How can we fix LLM facilitation? We investigate whether aggregating all currently available datasets can support either the training of classifiers or the use of out-of-the-box LLMs for a simpler and more commonly studied task in the literature: predicting whether a facilitator would intervene, rather than whether a facilitative comment is needed (§5). Our results show that LLMs are less reliable than ModernBert (Warner et al., 2025) classifiers, although their performance depends heavily on the model used (Finding 4).

Overall, our study offers the following contributions: We bridge differences between prior work on (i) what a facilitation intervention is, (ii) oral and written discussions (1), by compiling the largest standardized facilitation dataset so far (2). Through the execution of an extensive survey performed by experts on facilitation (3), we show that the key difference between humans and LLMs rests in how frequently they deem facilitation to be necessary (4). We are the first to evaluate LLMs for facilitation timing prediction, finding that they are wildly inconsistent while EncOn models offer a more reliable alternative, though both approaches remain constrained by the limited current understanding of facilitation within NLP (5).

Ultimately, achieving reliable automated facilitation requires moving beyond surface signals and using comprehensive, human-informed data that captures the true complexity of human facilitation. We release the code for the creation of our dataset and the experiments presented in this paper under a GPLv3-or-later license.[1]

2 Related Work and Background

2.1 What are facilitative interventions?

Facilitation (or conversational moderation) refers to the active participation of people in order to improve discussions (Korre et al., 2025). Relevant work assumes that facilitation is conducted by specially trained staff (Schroeder et al., 2024; Park et al., 2012; Chen et al., 2025), or users appointed by the community (Schaffner et al., 2024). Studying relevant literature on facilitation yields multiple definitions for what a facilitative intervention is, which can be categorized to three main definitions:

Definition 1 (Escalation)**.** A facilitator needs to intervene at the precise point at which, without facilitation, the discussion would derail or escalate.

[1] Dataset: https://anonymous.4open.science/r/facilitation-dataset-37DF, Experiments: https://anonymous.4open.science/r/interventions-6053, Survey Data: https://anonymous.4open.science/r/survey-data-C43E

Escalation is usually measured as the point where the participants request arbitration (e.g., escalate the matter to a site moderator in Wikipedia (De Kock and Vlachos, 2021; Chang and Danescu-Niculescu-Mizil, 2019)). This definition is quite appealing for NLP researchers as the labels are easy to acquire since a ground truth exists, and the existence of a single, largely unambiguous ground-truth is ideal for the training and evaluation of any Machine Learning (ML)-based classifier. However, a facilitator has multiple duties that do not directly relate to preventing escalation such as summarizing discussions (Small et al., 2023), managing participation (Chen et al., 2025; Schroeder et al., 2024), and providing important information for new members (Park et al., 2012). Furthermore, an unambiguous signal that a discussion has escalated is not often provided by platforms (Table 2). Thus, this definition is not considered in this study, but is included in our dataset (§3).

Definition 2 (Professional facilitation)**.** Facilitation should occur at the point where a professional facilitator would have intervened.

In current literature, it is generally accepted that all comments made by a facilitator should be considered facilitative interventions; for instance, only these comments are assumed to follow some facilitative strategy (Falk et al., 2021; Chen et al., 2025; Schroeder et al., 2024). Defining facilitative comments this way is simple and intuitive, but can only be used in domains where (1) professional facilitators exist and (2) they can be identified. These conditions exclude the vast majority of online discussions, as very few web-crawled datasets include such information.[2]

Definition 3 (Human judgment)**.** Facilitation should only occur when the majority agree that it is necessary.

There is an argument to be made that professional facilitators can make more accurate judgments due to expertise. However, in reality most facilitation is handled by either volunteers elevated to the position by their community (Schaffner et al., 2024) or by the users themselves in an ad-hoc fashion (Falk et al., 2024). In this case, the task is no longer about detecting or predicting whether *a facilitator would speak*, but rather the much more difficult and subjective task of whether the comment is *facilitative in nature*.

[2]See Table 2, where the largest datasets such as WikiConv (Hua et al., 2018) do not contain such information.

Name	Text	D1	D2	D3
Joe	Thank you for sharing. So I am Joe from Chicago originally...	×	×	×
Marie	Thank you Joe.	×	✓	?
Peter	Yeah. So I'll kind of go through it again. I am Peter...	×	×	×
Marie	Thank you, Peter. So I think we have another participant on the line...	×	✓	✓

Table 1: Example text where facilitation interventions are identified according to each definition (D1-3). *Marie* is a facilitator. We assume the discussion does not derail. The second comment *may* be facilitative according to Def. 3, which depends on the consensus of participants.

What is the difference? Table 1 shows an example where the same discussion is annotated following the three definitions. Def. 1 is reliant on the outcome of the discussion, and would miss examples where successful interventions led to a smooth discussion. Def. 2 would give more fine-grained information, although we run the risk of classifying non-facilitative comments as facilitative (and sometimes the opposite (Falk et al., 2024)). Finally, Def. 3 can provide us with fine-grained information, although ambiguous cases remain. It is worth noting that these definitions are not mutually exclusive, as demonstrated in Table 1.

2.2 Determining when to facilitate

Using models The only work to our knowledge that has explicitly attempted to predict facilitative interventions (Falk et al., 2021) uses EncOn classifiers on a single dataset–Cornell eRulemaking Initative (CeRI) (Park et al., 2012)–with modest performance. Indeed, these classifiers have proven competent in the context of escalation prediction (Chang and Danescu-Niculescu-Mizil, 2019; Yuan and Singh, 2023), and facilitative tactics detection (Chen et al., 2025; Schroeder et al., 2024). Recent work (Schroeder et al., 2024; Chen et al., 2025) uses LLM-as-a-judge models to annotate data in the field of facilitation, although recent research suggests that fully-automated facilitation is still out of our reach (Korre et al., 2025).

Using humans Expert human facilitators are frequently used in research (Chen et al., 2025; Park et al., 2012; Schroeder et al., 2024), but this is not a scalable process due to costs (Rossi et al., 2024).

Ignoring it Most platforms hold users responsible for maintaining the quality of discussion and the enforcement of community and platform rules, especially in jurisdictions where the platforms themselves are largely absolved of the responsibility of posted content, such as the U.S. (Seering, 2020; Schaffner et al., 2024). In some cases, the users are responsible for just flagging and reporting content (Seering, 2020), while platforms like Reddit use volunteer moderators, who also function as community leaders and facilitators (Seering, 2020; Falk et al., 2024), raising ethical questions about "Moderation as Free Labor" (Matias, 2019).

3 PEFK: A large, multi-domain facilitation dataset

Since there is currently no dataset that explicitly concerns itself with facilitation interventions, we construct the "Prosocial and Effective Facilitation in Konversations (PEFK)" dataset, a standardized, multi-domain compilation of the facilitation-related datasets listed by Korre et al. (2025) (see Table 2). PEFK contains data from both online forums and oral discussions, and standardizes metadata, metrics, and labeling schemes according to a simple, unified schema, making it suitable as a benchmark on facilitator interventions. More information on preprocessing steps and dataset schema can be found in App. B.

PEFK is designed to facilitate exploration and analysis of facilitative interventions under all three definitions described in §2.1. Specifically, we use a subset of the data that includes professional facilitation annotations to train our models according to Def. 2 in §5 (see App. B.3; Fig. 5). We also use a subset of PEFK to survey human facilitative patterns according to Def. 3. Finally, we include escalation and toxicity, as used in Def. 1, although this information is not utilized in this study (see §2.1).

4 Do humans and LLMs intervene at similar circumstances?

To support a systematic investigation of facilitative behavior under Def. 3, we conduct a survey on PEFK to analyze facilitation timing patterns between humans and LLMs. Rather than establishing a ground truth through Inter-Annotator Agreement (IAA), we adopt a survey-based approach, since prior work has shown that obtaining reliable annotations for similar tasks is highly challenging, particularly in newly explored settings such as ours (see "Survey Procedure").

4.1 Facilitative survey

Labels To avoid turning facilitation into a moderation task (i.e., using reactive responses after unwanted comments/utterances), we introduce two types of interventions: positive and negative reinforcement. *Positive reinforcement*, aims to encourage or reward constructive behavior (e.g., expressing appreciation or affirming contributions), while *negative reinforcement*, aims to discourage undesirable behavior or guide interlocutors back to productive discourse (e.g., addressing incivility or off-topic comments). This distinction represents a tool through which we ensured the survey participants did not default to content moderation, which we frequently observed during discussions with them, and during pilot survey rounds–it does not represent a theoretical contribution. Nevertheless, it provides an extra signal which we utilize in our analysis.[3] We adopt an ordinal scale approach, where the survey participants rate each dialogue for either type of reinforcement (positive or negative), as well as the possibility of no reinforcement using a scale from 1 to 5.[4]

Data We randomly split multi-party discussions from the PEFK dataset into small chunks containing a set amount of comments (6 comments per discussion), and select a random sample for the survey (1224 discussions–721 remained after the participants flagged some excerpts as malformed). This design aims to replicate a known limitation in online asynchronous facilitation scenarios; the facilitator is expected to decide on a course of action considering only the last parts of a discussion, as shown by several studies (Hewitt, 2003; Park et al., 2016; Kümpel and Unkel, 2021). In the case of oral discussions, there is evidence to suggest that the most recent comments are given relatively greater importance (Crano, 1977; Cockburn et al., 2017).

Survey Participants We recruited 11 experts on Computational Social Science with extensive experience with NLP tasks from sentiment analysis to persuasion strategy detection. We preferred ex-

[3]There is some theoretical grounding for using this approach is described in App. D.2. Similarly, Falk et al. (2021) use high and low Argument Quality for further analysis of facilitation prediction, although, unlike their approach, we obtain the explanatory signal directly from the survey participants, rather than established EncOn models.

[4]In order to transform the multiple ordinal labels into an easily manageable form for the purposes of analysis, we transform them into a single multiclass label schema as described in App. D.3.

Name	#Comments	#Discussions	Domain	Format	Fac.
WikiDisputes (2021)	96,320	8,727	Forum	Text	×
WikiTactics (WT) (2022)	3,850	213	Forum	Text	✓
WikiConv (2018)	17,806,373	4,486,983	Forum	Text	×
Conversations Gone Awry / CMV II (2019)	40,607	6,691	Forum	Text	×
Cornell eRulemaking Initative (CeRI) (2012)	3,700	132	Forum	Text	✓
User Moderation (UMOD) (2024)	2,000	1,000	Forum	Text	✓
Intelligence Squared (IQ2) (2016)	34,245	108	Formal debate	Oral	✓
Why-How-Who (WHoW) (2025)	25,542	178	Radio / TV	Oral	✓
Fora (2024)	39,438	262	Story sharing	Oral	✓

Table 2: The datasets used in PEFK include a large variety of domains, some crawled, and others as transcripts from oral discussions. All datasets include multi-participant discussions (besides UMOD which contains comment pairs). The **Fac.** column denotes whether information on which users are facilitators is available.

pert instead of crowdsourced participants as the task is highly technical, subjective, and mentally demanding. Additionally, crowdsourced participants have been shown to exhibit more bias in general (Snow et al., 2008; Waseem, 2016), which concerned us since we tried to avoid confusion with content moderation–a better known task. Our concerns were validated when preliminary discussions with participants revealed that they indeed considered their task to be content moderation.

The gender distribution was relatively balanced with 4 males and 7 females, while their ages had a median of 30 years. Before the commencement of the survey, the participants completed an agreement form in compliance with GDPR (European Union, 2016) ethical requirements and were compensated in accordance with national legislation. More information about the participant selection can be found in App. E.

Survey Procedure Unlike prior work that sought to establish consensus among facilitators (Falk et al., 2024; Schroeder et al., 2024; Chen et al., 2025), our goal is instead to explore facilitation patterns without presuming that such a consensus exists. Indeed, facilitation is a highly subjective task, and substantial disagreements persist even among experienced professionals operating within the same communities and under the same rulesets (Koshy et al., 2024); consensus remains elusive even in cases where the task is less ambiguous and the dataset more focused (Schroeder et al., 2024).[5]

Before the survey, the participants were given a training task where they categorized facilitative tactics according to established taxonomies on PEFK (see App. D.1) and given ample resources and time to familiarize themselves with online facilitation. We felt that training was a necessary step in order to disambiguate facilitation from content moderation after preliminary discussions with the participants. While it could in principle bias the participants, we attempted to control potential bias by strictly limiting the training scope to strategies only; there was no implicit or explicit input regarding the timing or flow of conversation (see App. D).

The main survey task required the participants to rate instances from PEFK by determining whether, at any point within a specific discussion chunk, reinforcement was needed, as well as what kind. The participants could optionally add rationales with which they could justify their rating. The full survey guidelines can be found in App. G. We ran two pilot surveys to clear up misunderstandings around the task and incorporate participant feedback, before executing the main task. Additionally, we conducted one-on-one interviews and weekly group meetings concurrently with the main task. More details can be found in App. E.

LLM Survey We use a variety of open-source LLMs (App. C.1) to ensure reproducibility. The prompts used were adopted from the guidelines provided to the experts–see App. H.

4.2 Human facilitation

Humans make cautious facilitators The main dilemma around facilitative timing is determining whether the facilitator prioritizes preventing potentially harmful speech, or letting the conversation continue naturally (Fig. 1). According to Fig. 2, most participants generally remain conservative–most choose not to intervene in a large majority of cases, although some (e.g., A3, A8, and A9) exhibit comparatively more proactive behav-

[5] During the survey, we inserted duplicated entries in secret to estimate consistency, finding that all-but-one participants remained 100% consistent in their judgments during the task–see App. D.5.

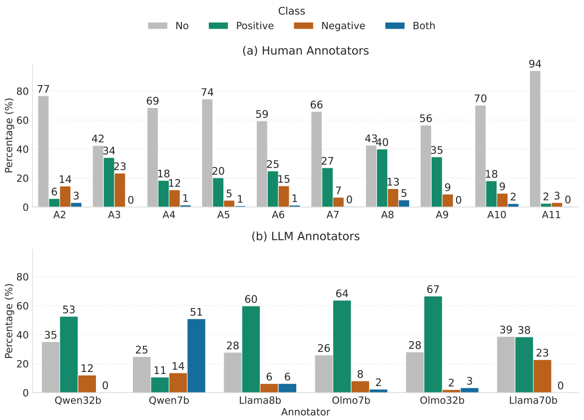

Figure 2: Distribution of all classes per participant. Human participants are either very cautious (A2, A4–A7, A10, A11) or somewhat cautious (A3, A8, A9). On the other hand, the LLMs (b) are significantly more eager to facilitate than humans (a).

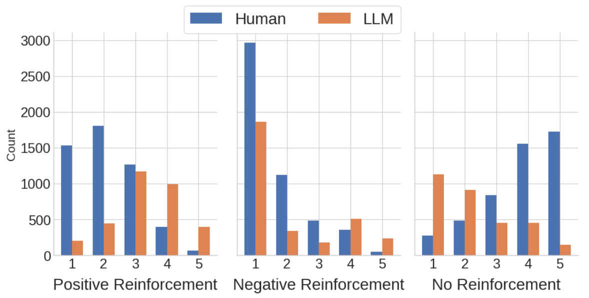

Figure 3: Certainty of judgments per facilitation label. LLMs (orange bars on the right) are much more confident in their judgments than humans (high frequencies on the extreme modes i.e., 1 and 5).

ior with higher rates of positive interventions.[6] Positive-intervention rates also appear to cluster around two broad ranges (approximately 20–30% and 35–40%), which is consistent with the tendencies of professional facilitators observed in the original dataset (see Fig. 5; App. B.2).[7]

Human facilitators may attend over different discussion spans To qualitatively understand the sources of divergence in human facilitative patterns, we analyzed a "worst-case" scenario: situations where half of the participants favored positive reinforcement and the other half favored negative reinforcement. As shown in Table 3, a specific discussion example illustrates this conflict. Participants justifying negative reinforcement cited the conver-

[6] One participant (A1) did not complete the task and was therefore not considered for the analysis.

[7] These observations are consistent across datasets–see App. D.4–reinforcing our motivation of transferring observations from oral datasets to online, written facilitation.

#	Content
1	Well nothing is really *inherently* bad. . .
2	Well political correctness is about avoiding offense. . .
4	Are you really arguing that WWI was unjust towards cisgendered men?
5	Not WWI itself, but that period, and I don't see what is so odd about it. . .
6	There were transgendered people in WWI. You still haven't justified how it was genuinely an injustice. . .

Table 3: Sections from the same discussion chunk. Participants facilitate differently depending on which section they decide is more important.

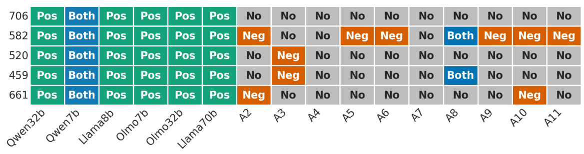

Figure 4: Top 5 discussion instances where the LLM majority vote disagrees with all human participants. The y-axis indicates the discussion instance index in the dataset. Labels follow the survey schema described in App. D.2: POS = Positive Reinforcement, NEG = Negative Reinforcement, BOTH = Both Positive and Negative Reinforcement, and NO = No Reinforcement.

sation's trajectory toward escalation, suggesting intervention was necessary to curb aggressive language. Conversely, the same participants noted that certain comments–particularly the first two–were examples of politeness and hedging, making them candidates for positive reinforcement.

Finding 1. *Most human participants tend to be either highly cautious or to balance proactivity with caution. Disagreements may arise because participants focus on different spans of the discussion.*

4.3 Do LLMs facilitate differently?

The facilitative style of LLMs differs significantly Fig. 3 shows that humans are less certain than LLMs when it comes to positive reinforcement but in both cases the values are concentrated in the middle part of the distribution. For negative reinforcement, both humans and LLMs show high uncertainty, while for no reinforcement there is a clear difference with humans being more certain that LLMs.

Why exactly do LLMs differ from humans? We identify cases where LLMs largely agreed among themselves but completely disagreed with all humans in order to highlight situations where model behavior diverges acutely from human judgment. In total, there are 36 cases where no human agrees with the majority voting of the LLMs. Fig. 4

shows the top 5 extreme cases. Despite strong LLM consensus (agreement >= 0.66), all human participants labeled these comments differently.

Examining the five examples qualitatively, we see that all cases regard a narrative of the participants rather than a debate like discussion that could possibly lead to escalation.[8] Specifically, discussion 706 is about gender, labor market change, and education, where multiple speakers exchange viewpoints on shifting economic roles.[9] The excerpt contains a lot of anecdotal examples (e.g., references to one's workplace examples and "Aviation High School"), rather than decision-oriented, focusing on broad social interpretation. In general, the dialogue is very self-contained, with speakers building perspectives rather than seeking resolution or guidance. This narrative like structure lead to the participant to choose to not to intervene and reporting that discussions like these comprise "neutral comments". LLMs, on the other hand, being more prone to intervention, could interpret this neutrality as an opportunity for positive reinforcement. This is a hypothesis that should be verified with rationales by the participants in future research.

Finding 2. *LLMs are more intrusive and more consistent compared to humans, being eager to view neutral discussions as opportunities for positive reinforcement.*

5 Can we use existing datasets?

As discussed in §2.1, facilitation interventions can be defined in two ways; whether a comment is facilitative in nature (Def. 3), or whether a dedicated facilitator would participate at a given point in time (Def. 2). In this section, we investigate whether we can use available datasets to train and evaluate models using the latter definition. See App. F.1 for experiments focusing on the broader understanding of facilitation by LLMs and App. F.2 for ablation studies on both self-consistency and instruction-sensitivity.

Experimental setup Def. 2 and Def. 3 are radically different, and thus require completely separate methodologies (see §2). We frame the task as a classification problem, aiming to identify the last comment made by a non-facilitator before a facilitator intervenes. To this end, we evaluate six open-source LLMs (App. C.1) and train a Modern-

Model	Pr.	Rec.	F1$_p$	F1$_n$	Sup.
CeRI					
~~LLaMa70B~~	~~0.152~~	~~0.082~~	~~0.107~~	~~0.783~~	~~335~~
LLaMa8B	0.246	0.518	0.333	0.567	335
ModBert	0.319	0.490	**0.386**	**0.706**	192
~~OLMo32B~~	~~0.154~~	~~0.071~~	~~0.097~~	~~0.795~~	~~335~~
~~OLMo7B~~	~~0.251~~	~~0.988~~	~~0.401~~	~~0.000~~	~~335~~
~~Qwen32B~~	~~0.100~~	~~0.024~~	~~0.038~~	~~0.821~~	~~335~~
~~Qwen7B~~	~~0.182~~	~~0.118~~	~~0.143~~	~~0.774~~	~~335~~
All	0.201	0.327	0.215	0.635	335
WT					
LLaMa70B	0.241	0.433	0.310	0.310	336
~~LLaMa8B~~	~~0.328~~	~~0.842~~	~~0.472~~	~~0.074~~	~~336~~
ModBert	0.407	0.717	**0.520**	**0.613**	258
OLMo32B	0.292	0.466	0.359	0.465	331
~~OLMo7B~~	~~0.348~~	~~0.958~~	~~0.511~~	~~0.009~~	~~336~~
Qwen32B	0.267	0.258	0.263	0.601	336
Qwen7B	0.264	0.392	0.315	0.455	336
All	0.307	0.581	0.393	0.361	336
Fora					
~~LLaMa70B~~	~~0.333~~	~~0.058~~	~~0.099~~	~~0.760~~	~~382~~
LLaMa8B	0.386	0.584	0.465	0.562	382
ModBert	0.409	0.631	**0.497**	0.599	2764
~~OLMo32B~~	~~0.391~~	~~0.066~~	~~0.113~~	~~0.765~~	~~382~~
~~OLMo7B~~	~~0.358~~	~~0.993~~	~~0.526~~	~~0.008~~	~~382~~
~~Qwen32B~~	~~0.438~~	~~0.051~~	~~0.092~~	~~0.773~~	~~382~~
Qwen7B	0.412	0.102	0.164	**0.759**	382
All	0.390	0.355	0.279	0.604	382
IQ2					
LLaMa70B	0.265	0.137	0.181	0.703	374
LLaMa8B	0.318	0.695	0.436	0.290	374
ModBert	0.399	0.634	**0.490**	0.572	2393
~~OLMo32B~~	~~0.273~~	~~0.092~~	~~0.137~~	~~0.736~~	~~374~~
~~OLMo7B~~	~~0.345~~	~~0.977~~	~~0.510~~	~~0.000~~	~~374~~
~~Qwen32B~~	~~0.167~~	~~0.038~~	~~0.062~~	~~0.743~~	~~374~~
Qwen7B	0.369	0.290	0.325	**0.693**	374
All	0.305	0.409	0.306	0.534	374
WHoW					
LLaMa70B	0.312	0.159	0.211	0.721	364
LLaMa8B	0.349	0.778	0.482	0.343	364
ModBert	0.418	0.626	**0.502**	0.597	1816
~~OLMo32B~~	~~0.455~~	~~0.079~~	~~0.135~~	~~0.779~~	~~364~~
~~OLMo7B~~	~~0.349~~	~~1.000~~	~~0.517~~	~~0.025~~	~~364~~
~~Qwen32B~~	~~0.296~~	~~0.063~~	~~0.105~~	~~0.762~~	~~364~~
Qwen7B	0.370	0.294	0.327	**0.697**	364
All	0.364	0.428	0.326	0.561	364
All datasets					
LLaMa70B	0.253	0.197	0.194	0.659	2000
LLaMa8B	0.313	0.677	0.425	0.388	2000
OLMo32B	0.285	0.149	0.162	**0.716**	1994
ModBert	0.400	0.555	**0.478**	0.609	2000
~~OLMo7B~~	~~0.316~~	~~0.986~~	~~0.476~~	~~0.007~~	~~2000~~
~~Qwen32B~~	~~0.236~~	~~0.105~~	~~0.121~~	~~0.728~~	~~2000~~
Qwen7B	0.296	0.229	0.242	0.686	2000
All	0.300	0.485	0.300	0.542	2000

Table 4: Classifier performance for facilitator prediction using positive-class precision, recall and F1 (F1$_p$) and negative-class F1 (F1$_n$). Rows are struck through when either F1$_p$ or F1$_n$ collapses (< 0.15). Best non-collapsed F1$_p$ and F1$_n$ values for each dataset split are highlighted in bold. "All" rows report macro averages across models.

[8] Unfortunately, we cannot rely on the LLMs for explanations post-hoc, nor use Chain of Thought (CoT) prompting to extract rationales, since both produce non-representative and deceptive explanations (Sanwal, 2025; Barez et al., 2025; Turpin et al., 2023). How the models arrived to these conclusions must therefore be evaluated from our own observations.

[9] Since each discussion span can involve as many as 30,000 characters, we refer the reader to the supplementary material for context.

Bert classifier (Warner et al., 2025).[10] Participant names are redacted, and each instance includes the three preceding comments as context. We use a 60–20–20 split for training the classifier, and choose 2,000 randomly sampled instances from the same test set for the LLM evaluation, evenly distributed across the sub-datasets of PEFK. We use a single Quadro RTX 6000 GPU for the classifier and two GPUs for the LLMs, collectively requiring about 72 hours of computation; see further details in App. C.2.

Deciding when NOT to intervene is easy. Table 4 shows the results for the facilitator intervention prediction task. We find that while most models are adept at identifying cases where facilitators did not intervene (large $F1_n$), they are generally much less successful at identifying when they *would* (poor $F1_p$) mirroring the fact that humans are more certain of when not to intervene (§4.2).

Finding 3. *Deciding when not to facilitate is easy both for humans and LLMs. Deciding when to facilitate, is not.*

Do LLMs over or under-intervene? LLM behavior in facilitation is polarized: some models exhibit a pervasive tendency to constantly facilitate (reflected by extremely low $F1_n$), a pattern documented in prior work (Tsirmpas et al., 2026). Conversely, we observe that other models rarely intervene (evidenced by extremely low $F1_p$–both crossed out in Table 4), a behavior that has been largely overlooked in the existing literature. We also note that discussion domain systematically influences LLMs but not humans in this regard–see App. F.5.

Are EncOn models the solution? We find that the ModernBert classifiers can generally compete with LLMs in this task, and seem much more consistent, outperforming them in three out of the five datasets, and scoring the highest $F1_p$ scores across all datasets.[11] We posit that overall, these classifiers may be a much more scalable, reliable and viable solution compared to LLMs for the task, even though they have been overlooked in current literature.[12]

Finding 4. *Out-of-the-box LLMs are generally not consistent in deciding when to facilitate, in contrast to traditional EncOn classifiers.*

6 Conclusion

We investigated how the timing of humans and LLMs differs in facilitative interventions. By conducting a literature review, we isolated three main definitions of facilitative interventions. We then conducted an survey task using facilitative social science experts on a standardized, aggregated corpus of all relevant facilitation datasets, finding that human facilitators tend to adopt a cautious approach, are more confident when deciding not to intervene than when deciding to intervene, and may disagree because they focus on different spans of the discussion. We are the first to conduct extensive experiments on facilitative intervention prediction using LLMs, finding that they tend to over-intervene, skew towards positive reinforcement when the situation is ambiguous, and remain inconsistent across facilitation decisions, while simpler ModernBERT classifiers trained on existing datasets can outperform them.

Despite optimism that LLMs would revolutionize facilitation, our findings suggest that progress may be incremental, not transformative–LLMs can not meaningfully facilitate until we move beyond limited datasets. Previous approaches–such as focusing on escalation or general moderation–provided large, scalable data, but only captured surface-level behaviors. They failed to give us a true understanding of how human facilitators operate, a complex and open question that can not be answered without annotated data that captures genuine facilitative dynamics.

Limitations

This work is not without limitations. With additional resources, we could have incorporated a larger survey or, given that we understood the task's subjectivity more adequately, even annotation schemata (e.g., sentence-by-sentence, apart from the excerpt) and explored a wider range of prompting settings. Also the opposite is true; a possible criticism of our decision is that we do not pro-

[10] Due to substantial differences in text format, we train separate models for oral and written datasets (App. B.3); an ablation study using a single model across all datasets shows reduced performance (App. F.3).

[11] There technically exists ways of boosting the performance of LLM classifiers, but this is outside the scope of this work–see App. C.1.

[12] Another notable advantage of these models is that we can tune the precision-recall tradeoff by adjusting the probability threshold over which the model determines that an intervention should happen, thus configuring how proactive or cautious they are–see App. F.4.

vide the full discussion thread, which in some cases could be extremely large and impractical to include in its entirety. Therefore, this organic limitation led to us to choose a middle ground, that is providing excerpts of the discussions. Facilitation decisions in real-world settings are often shaped by long-term interaction dynamics, participant history, community norms, and evolving conversational context, which may not be fully captured in the excerpts.

Furthermore, because providing rationales was optional for participants, we collected very few, resulting in the loss of a potentially valuable resource for interpretation–this information is valuable and should be incorporated in future tasks concerning facilitation. We only realized their usefulness during the analysis of our results, although it has to be acknowledged that making them mandatory would have significantly added to the workload of the participants.

Moreover, the intervention strategies examined here are only two: positive and negative reinforcement. Ideally, the task should be conducted without the use of different intervention labels or very fine-grained labels, as prior work has shown that different domains call for different facilitative tactic taxonomies (Park et al., 2012; Chen et al., 2025; Schroeder et al., 2024). The decision of including the distinction between negative and positive reinforcement was motivated by the fact that the participants were over-focusing on the moderation part during the refinement stage of the task. We also acknowledge that different discussion domains may favor different facilitative norms, which may have impacted the performance of our LLM facilitative classifiers. However, these domain-specific norms have not been explored in the literature, and the elicitation of optimal prompts is not our primary research objective, and thus considered outside of the scope of this work.

Additionally, although early work attempted to estimate interventions in terms of external metrics such as toxicity (Hua et al., 2018; Chang and Danescu-Niculescu-Mizil, 2019; Yuan and Singh, 2023), recent work has shown that facilitators may have different objectives depending on the form of discussion, which may require the use of different evaluation metrics for each domain (Korre et al., 2025). Since our work attempts to generalize interventions across multiple domains, we can not rely on external signals, but only on whether an intervention would (or did) occur.

Finally, the datasets included in PEFK are predominantly English-language and drawn from specific online or institutional discussion settings. As a result, the findings may not generalize to multilingual environments and culturally distinct moderation norms.

Ethical Considerations

All experiments and data have passed through official approval and were overseen by an Ethics Committee. The participants were selected via formal recruitment methods based on familiarity with the task and field, and were compensated fairly with a monthly salary of 1000€.[13] They signed informed consent forms before beginning the survey, were informed and consented to being exposed to potentially offensive or inappropriate content, and were allowed to skip uncomfortable tasks at any point, with no required justification. All necessary data were anonymized both during data collection and in the final, presented artifacts, while non-essential information was destroyed with the conclusion of the survey in accordance with GDPR (European Union, 2016) and the EU AI Act (European Union, 2024). Participant demographic data was not collected, since it was deemed unncessary for the purposes of this study.

In order to help research in the field of facilitation, we openly release the code that constructs PEFK under a GPLv3 license. All datasets used in PEFK are obtained from the sources provided by the respective authors and are consistent with intended use (academic research). We do not distribute the corpus separately, and thus no additional data license is applicable. The licenses of the original distribution apply. The Fora dataset (Schroeder et al., 2024) is not publicly available because it may contain personally identifying information; researchers should contact the authors of Fora for access. Finally, we use exclusively open-source models, libraries and LLMs, in order to ensure reproducibility, protect user data, and comply with relevant regulations (see App. C.1).

We used LLMs for some orthographic and grammatical support throughout the document, as well as for code generation in some instances. All outputs have been verified by the authors.

[13] Specific information on posted requirements and the recruitment process will be made available after review, since they would compromise anonymity.

References

Fazl Barez, Tung-Yu Wu, Iván Arcuschin, Michael Lan, Vincent Wang, Noah Siegel, Nicolas Collignon, Clement Neo, Isabelle Lee, Alasdair Paren, and 1 others. 2025. Chain-of-thought is not explainability. *Preprint, alphaXiv*, page v1.

James Bisbee, Joshua D. Clinton, Cassy Dorff, Brenton Kenkel, and Jennifer M. Larson. 2024. Synthetic replacements for human survey data? the perils of large language models. *Political Analysis*, 32(4):401–416.

Jason W. Burton, Ezequiel Lopez-Lopez, Shahar Hechtlinger, Zoe Rahwan, Samuel Aeschbach, Michiel A. Bakker, Joshua A. Becker, Aleks Berditchevskaia, Julian Berger, Levin Brinkmann, Lucie Flek, Stefan M. Herzog, Saffron Huang, Sayash Kapoor, Arvind Narayanan, Anne-Marie Nussberger, Taha Yasseri, Pietro Nickl, Abdullah Almaatouq, and 9 others. 2024. How large language models can reshape collective intelligence. *Nature Human Behaviour*, 8(9):1643–1655.

Jonathan P. Chang and Cristian Danescu-Niculescu-Mizil. 2019. Trouble on the horizon: Forecasting the derailment of online conversations as they develop. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 4743–4754, Hong Kong, China. Association for Computational Linguistics.

Ming-Bin Chen, Lea Frermann, and Jey Han Lau. 2025. WHoW: A cross-domain approach for analysing conversation moderation. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 2091–2126, Albuquerque, New Mexico. Association for Computational Linguistics.

Andy Cockburn, Philip Quinn, and Carl Gutwin. 2017. The effects of interaction sequencing on user experience and preference. *International Journal of Human-Computer Studies*, 108:89–104.

William D Crano. 1977. Primacy versus recency in retention of information and opinion change. *The Journal of Social Psychology*, 101(1):87–96.

Stefano Cresci, Amaury Trujillo, and Tiziano Fagni. 2022. Personalized interventions for online moderation. In *Proceedings of the 33rd ACM Conference on Hypertext and Social Media*, HT '22, page 248–251, New York, NY, USA. Association for Computing Machinery.

Christine De Kock, Tom Stafford, and Andreas Vlachos. 2022. How to disagree well: Investigating the dispute tactics used on wikipedia. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 3824–3837, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Christine De Kock and Andreas Vlachos. 2021. I beg to differ: A study of constructive disagreement in online conversations. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pages 2017–2027, Online. Association for Computational Linguistics.

European Union. 2016. Regulation (EU) 2016/679 of the European Parliament and of the Council.

European Union. 2024. Regulation (eu) 2024/1689 of the european parliament and of the council of 13 june 2024 laying down harmonised rules on artificial intelligence (artificial intelligence act). `https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng`. Official Journal of the European Union, L 2024/1689.

Neele Falk, Iman Jundi, Eva Maria Vecchi, and Gabriella Lapesa. 2021. Predicting moderation of deliberative arguments: Is argument quality the key? In *Proceedings of the 8th Workshop on Argument Mining*, pages 133–141, Punta Cana, Dominican Republic. Association for Computational Linguistics.

Neele Falk, Eva Vecchi, Iman Jundi, and Gabriella Lapesa. 2024. Moderation in the wild: Investigating user-driven moderation in online discussions. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 992–1013, St. Julian's, Malta. Association for Computational Linguistics.

Alvan R. Feinstein and Domenic V. Cicchetti. 1990. High agreement but low kappa: I. the problems of two paradoxes. *Journal of Clinical Epidemiology*, 43:543–549.

Rena Gao, Ming-Bin Chen, Lea Frermann, and Jey Han Lau. 2025. Moderation matters: Measuring conversational moderation impact in English as a second language group discussion. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 2070–2095, Vienna, Austria. Association for Computational Linguistics.

Jim Hewitt. 2003. How habitual online practices affect the development of asynchronous discussion threads. *Journal of Educational computing research*, 28(1):31–45.

Yiqing Hua, Cristian Danescu-Niculescu-Mizil, Dario Taraborelli, Nithum Thain, Jeffery Sorensen, and Lucas Dixon. 2018. Wikiconv: A corpus of the complete conversational history of a large online collaborative community. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2818–2823, Brussels, Belgium. Association for Computational Linguistics.

Katerina Korre, Dimitris Tsirmpas, Nikos Gkoumas, Emma Cabalé, Danai Myrtzani, Theodoros Evgeniou, Ion Androutsopoulos, and John Pavlopoulos.

2025. Evaluation and facilitation of online discussions in the LLM era: A survey. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 24443–24462, Suzhou, China. Association for Computational Linguistics.

Vinay Koshy, Frederick Choi, Yi-Shyuan Chiang, Hari Sundaram, Eshwar Chandrasekharan, and Karrie Karahalios. 2024. Venire: A machine learning-guided panel review system for community content moderation. *CoRR*, abs/2410.23448.

Anna Sophie Kümpel and Julian Unkel. 2021. (why) does comment presentation order matter for the effects of user comments? assessing the role of the availability heuristic and the bandwagon heuristic. *Communication Research Reports*, 38(4):217–228.

Jorge Nathan Matias. 2019. The civic labor of volunteer moderators online. *Social Media + Society*, 5.

Deokgun Park, Simranjit Sachar, Nicholas Diakopoulos, and Niklas Elmqvist. 2016. Supporting comment moderators in identifying high quality online news comments. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, pages 1114–1125.

Joonsuk Park, Sally Klingel, Claire Cardie, Mary Newhart, Cynthia Farina, and Joan-Josep Vallbé. 2012. Facilitative moderation for online participation in erulemaking. In *Proceedings of the 13th Annual International Conference on Digital Government Research*, dg.o '12, page 173–182, New York, NY, USA. Association for Computing Machinery.

Perspective API. 2026. Perspective api. Accessed: 2026-02-24.

Luca Rossi, Katherine Harrison, and Irina Shklovski. 2024. The problems of llm-generated data in social science research. *Sociologica*, 18(2):145–168.

Paul Röttger, Bertie Vidgen, Dirk Hovy, and Janet Pierrehumbert. 2022. Two contrasting data annotation paradigms for subjective NLP tasks. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 175–190, Seattle, United States. Association for Computational Linguistics.

Manish Sanwal. 2025. Layered chain-of-thought prompting for multi-agent llm systems: A comprehensive approach to explainable large language models. *arXiv preprint arXiv:2501.18645*.

Brennan Schaffner, Arjun Nitin Bhagoji, Siyuan Cheng, Jacqueline Mei, Jay L Shen, Grace Wang, Marshini Chetty, Nick Feamster, Genevieve Lakier, and Chenhao Tan. 2024. "community guidelines make this the best party on the internet": An in-depth study of online platforms' content moderation policies. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, CHI '24, New York, NY, USA. Association for Computing Machinery.

Hope Schroeder, Deb Roy, and Jad Kabbara. 2024. Fora: A corpus and framework for the study of facilitated dialogue. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13985–14001, Bangkok, Thailand. Association for Computational Linguistics.

Joseph Seering. 2020. Reconsidering self-moderation: the role of research in supporting community-based models for online content moderation. *Proc. ACM Hum.-Comput. Interact.*, 4(CSCW2).

Tsukasa Shiota, Takashi Yamamura, and Kazutaka Shimada. 2018. Analysis of facilitators' behaviors in multi-party conversations for constructing a digital facilitator system. In *Collaboration Technologies and Social Computing*, pages 145–158, Cham. Springer International Publishing.

Christopher T. Small, Ivan Vendrov, Esin Durmus, Hadjar Homaei, Elizabeth Barry, Julien Cornebise, Ted Suzman, Deep Ganguli, and Colin Megill. 2023. Opportunities and risks of llms for scalable deliberation with polis. *ArXiv*, abs/2306.11932.

Rion Snow, Brendan O'Connor, Daniel Jurafsky, and Andrew Ng. 2008. Cheap and fast – but is it good? evaluating non-expert annotations for natural language tasks. In *Proceedings of the 2008 Conference on Empirical Methods in Natural Language Processing*, pages 254–263, Honolulu, Hawaii. Association for Computational Linguistics.

Amaury Trujillo and Stefano Cresci. 2022. Make reddit great again: Assessing community effects of moderation interventions on r/the_donald. *Proc. ACM Hum.-Comput. Interact.*, 6(CSCW2).

Dimitris Tsirmpas, Ion Androutsopoulos, and John Pavlopoulos. 2026. Designing synthetic discussion generation systems: A case study for online facilitation. *Preprint*, arXiv:2503.16505.

Miles Turpin, Julian Michael, Ethan Perez, and Samuel Bowman. 2023. Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. *Advances in Neural Information Processing Systems*, 36:74952–74965.

Benjamin Warner, Antoine Chaffin, Benjamin Clavié, Orion Weller, Oskar Hallström, Said Taghadouini, Alexis Gallagher, Raja Biswas, Faisal Ladhak, Tom Aarsen, Griffin Thomas Adams, Jeremy Howard, and Iacopo Poli. 2025. Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2526–2547, Vienna, Austria. Association for Computational Linguistics.

Zeerak Waseem. 2016. Are you a racist or am I seeing things? annotator influence on hate speech detection on Twitter. In *Proceedings of the First Workshop on NLP and Computational Social Science*, pages 138–142, Austin, Texas. Association for Computational Linguistics.

Jiaqing Yuan and Munindar P. Singh. 2023. Conversation modeling to predict derailment. *Proceedings of the International AAAI Conference on Web and Social Media*, 17(1):926–935.

Justine Zhang, Ravi Kumar, Sujith Ravi, and Cristian Danescu-Niculescu-Mizil. 2016. Conversational flow in Oxford-style debates. In *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 136–141, San Diego, California. Association for Computational Linguistics.

A Acronyms

NLP Natural Language Processing

NLU Natural Language Understanding

ML Machine Learning

RQ Research Question

PEFK Prosocial and Effective Facilitation in Konversations

EncOn Encoder-Only

IAA Inter-Annotator Agreement

UMOD User Moderation

IQ2 Intelligence Squared

CeRI Cornell eRulemaking Initative

WHoW Why-How-Who

WT WikiTactics

CoT Chain of Thought

B Dataset details

B.1 Preprocessing

General

- We exclude comments with no text.
- We exclude discussions with less than two distinct participants.
- We exclude discussions which are common between wikitactics and wikiconv as well as wikidisputes and wikiconv.
  - There may be duplicate discussions between wikidisputes and wikitactics, but we allow them since they feature complementary information.

Wikiconv

- The Wikiconv corpus does not contain information about which user is a moderator/facilitator. Therefore, all comments relating to Wikiconv are tagged as non-moderators.
- Additionally, we follow the instructions of the original researchers (Hua et al., 2018), and select only discussions which have at least two comments by different users.
  - Wikipedia (thankfully) does not track users who log in with only an IP address (in the original dataset, their `user_id` is always set to 0 and their username is of the form `211.111.111.xxx`. We consider each such username to be a separate user.
  - Due to the size of the dataset, we have to partially load it during preprocessing. Thus, there is a small chance every 100,000 records that a discussion is marked as a false negative and a part of it gets discarded.
  - We only include English comments in the final dataset. We use a small, efficient library (`py3langid`) for language recognition, due to the large size of Wikiconv. Non-English comments are discarded before selecting valid discussions (see point above).

Wikitactics

- We infer facilitative actions by whether the comment belongs in any of the following categories:
  - Asking questions
  - Coordinating edits
  - Providing clarification
  - Suggesting a compromise
  - Contextualisation
- The above tactics are a subset of the Coordinative labels used in the WikiTactics paper. They were selected because they are not used necessarily on 1-1 discussions; they could reasonably be applied by third-party participants. Contrast them with other Coordinative labels such as "Conceding/recanting" and "I don't know".

Wikidisputes

- Since only 0.03% of the comments in the dataset are made by moderators, we mark the dataset as not supporting moderation.

Name	Type	Description
conv_id	string	The discussion's ID. Comments under the same discussion refer to the same discussion ID.
message_id	string	The message's (comment's) unique ID.
reply_to	string	The ID of the comment which the current comment responds to. nan if the comment does not respond to another comment (e.g., it's the Original Post (OP)).
user	string	Username or hash of the user that posted the comment
is_moderator	bool	Whether the user is a facilitator.
moderation_supported	bool	True if the moderation labels are directly computed from the original dataset
escalated	bool	A discussion-level measure denoting discussions which have been derailed
escalation_supported	bool	True if the escalation labels are directly computed from the original dataset
text	string	The contents of the comment
dataset	string	The dataset from which this comment originated from
notes	JSON	A dictionary holding notable dataset-specific information
toxicity	float	The "toxicity" score given to the comment by the Perspective API
severe_toxicity	float	The "severe toxicity" score given to the comment by the Perspective API

Table 5: The full set of columns used for each comment in the PEFK dataset.

UMOD

- Facilitative actions are marked as a gradient from 0 (no facilitation) to 1 (full facilitation). We adopt a threshold of 0.75 to consider an action as facilitative, with more than 50% participant agreement (measured as entropy in the original dataset).

CMV-AWRY2

- We mark a discussion as escalated when the derailment value (from the official dataset) is in the 60th upper percentile.
- We remove deleted ("[deleted]") comments.

B.2 Schema

Table 5 details the information available for each comment in the dataset. Some information, such as whether the person talking is a moderator / facilitator, is available only on some sub-datasets. Also note that the `is_moderator` column may hold different definitions of what we consider as moderator / facilitator. UMOD for instance considers facilitative comments made by users, while others such as Fora consider a comment facilitative if it was made by a dedicated facilitator.

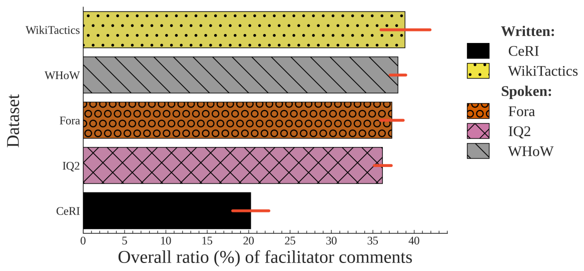

Figure 5: Percentage of comments made by professional facilitators (see Def. 2) for each of the sub-datasets of PEFK. Bars denote 95% confidence intervals. Despite the different kinds of facilitators in online platforms and various real-life-discussion domains, the overall rate of facilitation remains similar across datasets.

B.3 Exploratory Data Analysis

We explore some initial frequencies with regard to facilitator comments in PEFK, comparing its oral vs. written subsets, as well as vocabulary discrepancies.

Oral vs. written datasets in PEFK There currently exists a rift in facilitation research; while some previous work concerns itself with forums and online discussions (Park et al., 2012), others are exclusively focused on live, real-life, oral discussions (Schroeder et al., 2024; Chen et al., 2025).

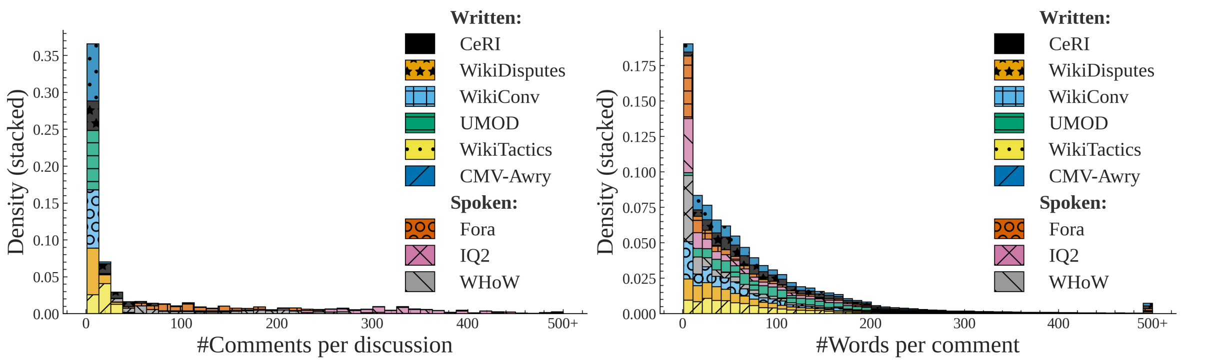

Figure 6: **Left:** Comments per discussion on the PEFK dataset. Note the significant increase in comments made in transcribed datasets such as Fora and IQ2. This deviation is explained by the fact that a single human utterance is regularly broken up to many comments in order to simulate interruptions or participants talking over each other. **Right:** Number of words per comment in all discussions in PEFK, grouped by original dataset. Comments across written datasets follow the same distribution, while oral datasets feature a significant number of comments with very few words.

This rift is an issue for facilitation research, since oral and online written discussions feature fundamentally different kinds of facilitation interventions, as shown by stark differences in proposed facilitation schemas between online forums (Park et al., 2012), and real-life discussions (Schroeder et al., 2024; Chen et al., 2025). Additionally, the data themselves are different; posts and comments are self-contained, while transcripts regularly feature interruptions, participants talking in parallel (see App. B.3), and linguistic features that are necessary for coordination (e.g., "Thank you.", "I agree."). It is worth noting that in our case, all written discussions are asynchronous (the participants may join and leave at any time), while all oral discussions are synchronous (the participants are all active at the same time).

Figure 6 shows some of the structural differences of oral and written datasets. oral datasets feature a significantly larger number of comments per discussion, since instances where the participants talk over each other, or interrupt the other are transcribed as separate comments. Additionally, oral datasets feature a large amount of comments with very few words, since a lot of utterances are short but necessary for social co-ordination (e.g., "Thank you", "Yes", "I agree"). Figure 5 shows the percentage of comments made by facilitators for each of our datasets. In almost all datasets, a third of the comments are made by facilitators.[14]

[14]We do not include the UMOD dataset, since there is no established ground-truth (see App. B.1).

Vocabulary discrepancies in PEFK datasets. We also the data to include only messages written by moderators, then cleaned the text by removing URLs, punctuation, and other noise. After preprocessing, we used a TF-IDF vectorization approach with n-grams (bigrams and trigrams) to extract phrases. Crucially, the TF-IDF model was trained on all datasets combined, so that common, generic phrases were downweighted while phrases that are more distinctive to each dataset were emphasized. We then computed the average TF-IDF score of each phrase within each dataset and selected the top-ranking ones as representative expressions of moderator language in that context. The results can be found in Figure 7.

The most important phrases in CeRI suggest a formal, policy-oriented environment with an emphasis on structured participation and regulatory discussion (e.g., references to "proposed rule" and "welcome regulations"). In *fora*, the language is more conversational and facilitative, with phrases such as "great thank", "feel free", and "awesome thank" indicating a more informal moderation style. The IQ2 dataset reflects the debate setting, with phrases like "ladies gentlemen" and "arguing motion". In UMOD, moderators engage more in argumentation and clarification, as seen in phrases like "change view", "missing point" reminiscing more facilitative settings and suggesting a focus on reasoning and discussion quality. The WHoW dataset appears similar to moderated discussions or panels, with phrases like "question sir" and "answer question" indicating turn-taking and audience interaction. Finally, WT shows a highly procedural

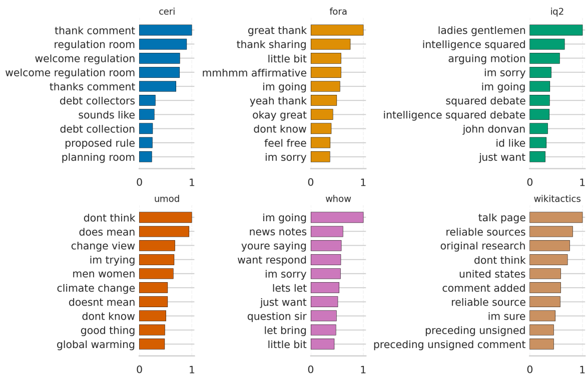

Figure 7: Most common moderator bigrams in PEFK per dataset. Function words are excluded.

and rule-based style, with phrases such as "reliable sources", "original research" and "talk page" reflecting the norms and policies of collaborative editing environments like Wikipedia.

C Models

C.1 LLMs

We use only open-source models, as proprietary models such as OpenAI's GPT-5 yield much less reproducible results (Bisbee et al., 2024). Furthermore, the use of open-source models requires much fewer resources, enabling research on this task with a less demanding budget, as well as scalable deployment in real-world settings (Tsirmpas et al., 2026). It is also worth noting that since a part of our dataset (Fora) is not licensed, we can not risk leaking the data contained within to proprietary models. Generally, the use of local models protects the personal data of the users contained in our datasets, ensures that our research can be replicated in domains where anonymity must be maintained, and renders our work compatible with existing legislation (European Union, 2024, 2016).

Proprietary LLMs such as the OpenAI GPT family may perform better, but cost several orders of magnitude more, and are thus not a realistic solution (Tsirmpas et al., 2026). While the performance of LLM classifiers could be reasonably increased by providing few-shot examples or finetuning, these approaches would significantly bias the model since no concrete definition of facilitation exists (Korre et al., 2025), and the examples would have to be different for each different discussion domain in PEFK–which goes against our motivation of discovering patterns between oral and written datasets.

Table 6 shows details on the LLMs used for this study. We specifically use 4-bit quantized versions from the `unsloth huggingface` repository.

Family	Version	#Params
`OLMo`	`2 (0325)`	32B
`Qwen`	`2.5`	32B
`LLaMa`	`3.3`	70B
`OLMo`	`3`	7B
`Qwen`	`2.5`	7B
`LLaMa`	`3.1`	8B

Table 6: LLMs used in this study.

C.2 Transformers

We construct our training and evaluation data from a unified corpus of threaded discussions, where each instance corresponds to a target comment augmented with its conversational context. For every target comment, we retrieve up to $K = 3$ preceding comments along the reply chain. Each comment is truncated to a maximum of 3000 characters to control sequence length. The input is formatted using XML-style tags, where context comments are wrapped with `<CTX>` and the target comment with `<TGT>`. The final sequence is formed by concatenating context comments in chronological order followed by the target comment. We consider binary classification tasks (e.g., `is_moderator`, `should_intervene`), with labels cast to floating point values.

We fine-tune a pretrained transformer encoder (`ModernModBert-large`) with an added classification head that outputs a single logit. The task is formulated as a single-label instance of multi-label classification, enabling the use of a sigmoid activation.

Training is conducted using the Hugging Face `Trainer` framework with several modifications. We optimize the model using binary cross-entropy with logits:

$$\mathcal{L} = \text{BCEWithLogitsLoss}(\mathbf{z}, \mathbf{y}; \text{pos\_weight}),$$

where $\mathbf{z}$ are the predicted logits and $\mathbf{y}$ are the ground-truth labels. To account for class imbalance, the positive class is weighted as. We freeze the base transformer parameters and train only the classification head in order to reduce computational cost and limit overfitting.

To improve computational efficiency, we employ a bucketed batching strategy that groups instances of similar length (approximated in characters) into batches of size 64. This reduces padding and improves throughput. Batches are shuffled at the bucket level at each epoch.

Training is performed for up to 80 epochs, with evaluation, logging, and checkpointing conducted once per epoch. The best model is selected based on validation loss. We apply early stopping with a warm-up phase: monitoring begins after one evaluation step, and training stops if validation loss does not improve by at least $\delta = 0.001$ for 6 consecutive evaluation steps. Model performance is evaluated using accuracy and binary F1 scores.

All experiments are implemented in PyTorch using the Hugging Face Transformers library. Data loading uses 4 worker processes, and GPU acceleration is utilized when available. We fix the random seed to 42 to ensure reproducibility.

Taxonomy & Tactic	**Type**
Fora	
Express appreciation	+
Express agreement or affirmation	+
Open invitation to participants	
Specific invitation to participant	
Model examples	
Follow-up question	
Make connections	
WHoW	
Informational	
Coordinative	-
Social	+
Probing	
Confronting	
Instruction	-
Interpretation	
Supplement	
CeRI	
Welcoming	+
Encouragement / appreciation	+
Maintaining civil discourse	-
Explaining why comment not removed	
Indicating irrelevant/off-point comments	-
Pointing to relevant information	
Pointing out effective commenting	+
Asking for more information/sources	
Asking for solutions/alternatives	
Encourage engagement with others	
Pose question to community	

Table 7: Facilitation taxonomies (bold) with associated tactics (normal text) and reinforcement type. +: positive reinforcement, -: negative reinforcement, empty if the tactic does not clearly correspond to any category.

D Human survey procedure

D.1 Facilitative training

Before the main survey on facilitative intervention, all participants were extensively trained on currently established facilitative tactics. This task aimed to familiarize the participants with what constitutes a facilitative intervention, as well as how professional facilitators operate in real-life scenarios.

Specifically, we split the participants into three groups, each of which was assigned a different taxonomy of facilitative interventions (Chen et al., 2025; Schroeder et al., 2024; Park et al., 2012). The participants were tasked with classifying 830 facilitated comments from a sample of continuous discussions in PEFK. The guidelines and results of this annotation are not shown in this study, but they are available upon request.

D.2 Positive vs. negative reinforcement

The distinction between positive/negative reinforcement is present to suggested facilitative tactics taxonomies (Table 2), including prior work such as Fora (Schroeder et al., 2024) WHoW (Chen et al., 2025), and CeRI (Park et al., 2012). We conservatively estimate whether the tactics described in each taxonomy correspond to our definitions of positive and negative reinforcement.

The facilitative taxonomy proposed in Fora does not include any tactics that could be unambiguously considered as negative reinforcement, which is expected, since facilitators deployed in the relevant study were supposed to only act as coordinators. Notably, several tactics may simultaneously exhibit both positive and negative characteristics across all taxonomies, depending on context and the exact implementation used by the facilitator. We therefore use these taxonomies only as guidance during annotation, and refer the reader to the original works for a more comprehensive description and analysis of facilitation strategies.

D.3 Label Transformation

To efficiently calculate intra-participant consistency, we transform the fine-grained, ordinal label per type adopting a multiclass schema. We define a threshold of >=3 and then using Table 8 we relabel the items.

Positive	Negative	Neither	Final
✗	✗	✗	None (0)
✓	✗	✗	Positive (1)
✗	✓	✗	Negative (2)
✗	✗	✓	None (0)
✓	✓	✗	Both (3)
✓	✗	✓	Positive (1)
✗	✓	✓	Negative (2)
✓	✓	✓	Both (3)

Table 8: Reinforcement label transformations.

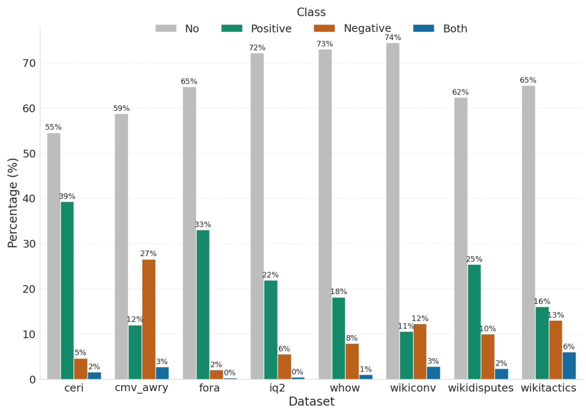

Figure 8: Label distributions of human participants per dataset.

D.4 Are facilitative tendencies dataset-dependent?

Facilitative tendencies generally follow a conservative pattern, with no reinforcement selected most often. Despite the differences between the datasets (forums, debates, and radio podcasts), we observe similar trends in the choice between positive and negative reinforcement, with positive reinforcement being more common overall. Two exceptions are cmv_awry and wikiconv, where negative interventions are selected more often than positive ones, by 15% and 1%, respectively (Fig. 8).

D.5 Participant consistency

To evaluate the consistency of individual participants, we measured intra-participant agreement using purposefully repeated instances of identical discussions.

Agreement was computed as a strict match in the labeling of the same instance (duplicated pair). For each duplicated pair, agreement was assigned a value of 1 if both labels were identical and 0 otherwise in order to calculate a simple pairwise agreement. Hence, the final intra-participant agreement for each participant was then calculated as the mean of these binary agreement scores. Since our classes are heavily skewed and imbalanced, traditional agreement metrics tend to collapse, showing extremely low scores even for almost-100% agreement (Röttger et al., 2022; Feinstein and Cicchetti, 1990). For the purposes of simplicity, we thus run our analysis in this part with simple percentage agreement.

participant	Agreement (%)
A2	100.00
A3	66.67
A4	88.89
A5	100.00
A6	100.00
A7	100.00
A8	100.00
A9	100.00
A10	100.00
A11	100.00

Table 9: Intra-participant percentage agreement (consistency) over all subdatasets of PEFK.

We report results in Table 9. Most participants demonstrated near-perfect consistency, with several achieving 100% agreement. Participant A3 is the only participant who showed reduced consistency (66.67%), suggesting potential variability in interpretation or labeling behavior.

E Participants Selection and Coordination

Selection Participant selection required a response to the public call, followed by a round of interviews to verify that candidates fulfilled the criteria for the role. All successful participants held a BA in Languages, Communication, or History, and an MSc in either Digital Humanities or Language Technology. They also had experience with NLP annotation tasks, such as sentiment analysis and toxicity detection. We would also like to highlight that controls were conducted before the survey (hiring), during the survey (training and discussion), and after the survey (removal of outlier or inconsistent participants), to ensure the highest quality of results.

Coordination The coordination of the project was largely based on weekly meetings that allowed the integration of participant feedback and potential

Ph.	Dates	Activity
1	Dec	Recruitment
2	Dec–Jan	Training on facilitation strategies
3	Jan 1–9	Onboarding
4	Jan 9–15	Pilot
5	Mid Jan–Mid Feb	Main experiment and discussion
6	End Feb–Mid Mar	Main experiment completion
1–6	Weekly	Discussion & feedback rounds

Table 10: Coordination timeline.

changes to our survey methodology. An overview of the project timeline can be found in Table 10. Before the commencement of the main survey, two crucial steps were carried out. The first was familiarizing participants with facilitation tactics; more information can be found in Section D.1. The second was a pilot round with ∼160 instances, which allowed us to better understand the challenges of the task. For example, one major concern was time. Depending on instance length, labeling took between 2 and 5 minutes. This led us to break the task into smaller subtasks, spread across almost 3 months (see Phases 5 and 6 in Table 10), with annotators labeling about 400 instances per round.

F Additional Experiments

F.1 Can LLMs understand what facilitation is?

One way of establishing whether the models can understand the concept of facilitation is exploring whether these models can distinguish between facilitative and non-facilitative comments in the first place. Their performance in this task should give us an indication of how well-defined the task is in relevant NLP datasets. We thus explore *"facilitator detection"* a Natural Language Understanding (NLU) task which aims to identify whether a comment was made by a facilitator (Def. 2). This task has been explored exactly once by Shiota et al. (2018), although the authors used hand-crafted features instead of relying exclusively on the language used by the facilitators, and used only role-play focused scenarios instead of real-life facilitation ones.

Table 14 presents the performance of these same models in the task of facilitation detection. While we observe instances of strong performance–such as Qwen32B achieving high scores on CeRI ($F1_p$=0.904, $F1_n$=0.971)–the models generally do not score reliably across diverse datasets (e.g., LLaMa70B in WT yields $F1_p$=0.500). This suggests that the fundamental cause of underperformance is less a confusion with the traditional moderator role and more a structural issue with the datasets themselves. Prior work utilizing traditional ML (Shiota et al., 2018), LLMs, and EncOn classifiers shows similar challenges on other tasks (Chen et al., 2025), indicating that this finding is not merely a result of our specific labeling scheme or dataset selection.[15] Thus, significant, generalizable improvements can only be realized by incorporating larger professional facilitation datasets, where interventions can be *reliably defined and distinguished* from non-facilitative comments, or by developing larger-scale corpora that contain annotated *facilitative comments* (Def. 3)—an extremely difficult task given the inherent subjectivity of the task (§4).

F.2 LLM sensitivity ablations

We run an ablation study using five instruction prompt variants on a 20% sample of the dataset. We generated these variants via a local Gemma-4 model (specifically, the https://huggingface.co/unsloth/gemma-4-E4B-it-GGUF variant, generated using the TextGen open-source software—https://github.com/oobabooga/textgen) in order to avoid researcher bias, and verified the resulting output. Two of these variants include specific instructions warning against over-intervention ("cautious 1,2"). The pairwise agreement scores for four models can be found in Table 11 and the instruction variants in App. H.2. We find that LLM judgments seem to remain broadly consistent, even with large changes to the LLM input for the specific task. Some inconsistencies do exist, as is the case with all LLM-related tasks (Bisbee et al., 2024), and some models (such as Qwen-7B) may be more sensitive than others.

We also executed a similar ablation experiment as the one above, but instead of testing five different prompts, the models were given the original instruction prompt five times. The pairwise agreement scores for each run can be found in Table 12. It appears that our attempts to restrict stochasticity within the experiment (e.g., the standardization of python, numpy and pytorch seeds), as well as the

[15] Schroeder et al. (2024) feature F1 scores in the 0.7-0.9 range, significantly outperforming our own models. However, we were not able to replicate these results given their EncOn model setup. While the authors acknowledged our requests for clarifications in their training and evaluation setup, we have not received any.

Run A	Run B	LLaMa-70B	Qwen-32B	OLMo-32B	OLMo-7B	Qwen-7B	Llama-8B
cautious1	cautious2	0.975	0.978	0.888	0.972	0.966	0.972
cautious1	concise	0.880	0.771	0.771	0.808	0.756	0.776
cautious1	descriptive	0.759	0.870	0.739	0.760	0.587	0.741
cautious1	role-play	0.945	0.664	0.799	0.689	0.775	0.807
cautious2	concise	0.867	0.784	0.898	0.807	0.760	0.767
cautious2	descriptive	0.785	0.863	0.797	0.756	0.585	0.745
cautious2	role-play	0.941	0.662	0.790	0.695	0.790	0.794
concise	descriptive	0.708	0.706	0.869	0.859	0.816	0.737
concise	role-play	0.874	0.771	0.830	0.785	0.598	0.838
descriptive	role-play	0.821	0.619	0.799	0.815	0.423	0.744

Table 11: Pairwise similarity scores across prompt variants for different language models.

nature of the task, lead to deterministic outputs by the LLMs on the evaluated sample.

F.3 One model to rule them all?

Since comments in the written and oral datasets are fundamentally different (§B.3), we would be tempted to train different models for each. This decision is not obvious however, since it is likely that the model can internally distinguish between oral and written speech. If that occurs, the differences between the texts should not negatively affect the model; in fact, training the model on all datasets could allow it to transfer knowledge between the two text modalities. Table 13 shows the performance of the same model when trained only on oral and written datasets, compared to when trained on the all datasets at once. We find that the models perform better when trained on different splits.

F.4 Decision thresholds

An advantage of using EncOn classifiers instead of LLMs is that we can quantitatively alter the performance of the classifier by tuning the thresholds required for a comment to trigger an intervention. By making the model more conservative (increasing the threshold), we could hope that we can avoid unnecessary interventions, while making it less conservative (decreasing the threshold) we could make it more proactive.

Fig. 9 shows that the classifier has an inherently low precision score which is largely invariant to the selected threshold. On the other hand, recall falls off rapidly, which incentivises us to use a more liberal approach.

F.5 Is facilitator behavior dependent on domain?

Fig. 10 shows the ratio of facilitative interventions for human and LLM survey participants.[16] Human facilitators are generally invariant to the discussion domain. However, most LLMs tend to under-facilitate in non-forum domains, and over-facilitate in WT, the only dataset featuring a traditional forum structure. A few over-facilitate in general, as seen in Table 4.

Why do we observe this discrepancy between domains? It could be caused by some dataset-specific linguistic markers or latent pattern systematically influencing the judgments of LLM facilitators. Indeed, WT is the only dataset considered in this study specifically containing escalating discussions, which likely triggers LLM interventions. Human facilitators do not seem to be influenced by the escalating discussions however (Fig. 10). This discrepancy could be attributed to LLMs being hypersensitive to unsafe speech because of their extensive alignment procedures.

An alternative explanation would posit that LLMs may not understand the existence of facilitative interventions in domains other than traditional online forums. Under this assumption the LLMs could abstain from intervening not because they "calculate" that a facilitation intervention would be unnecessary, but rather because they have not been exposed to facilitative interventions during their training. In any case, this assumption does not seem to be hold; Table 14 shows that the LLMs are reasonably competent at detecting facilitators in multiple domains.

[16] Technically, LLMs do not take surveys, they execute a task phrased as a survey for humans.

Run A	Run B	LLaMa-70B	Qwen-32B	OLMo-32B	OLMo-7B	Qwen-7B	Llama-8B
1	2	1.000	1.000	1.000	1.000	1.000	1.000
1	3	1.000	1.000	1.000	1.000	1.000	1.000
1	4	1.000	1.000	1.000	1.000	1.000	1.000
1	5	1.000	1.000	1.000	1.000	1.000	1.000
2	3	1.000	1.000	1.000	1.000	1.000	1.000
2	4	1.000	1.000	1.000	1.000	1.000	1.000
2	5	1.000	1.000	1.000	1.000	1.000	1.000
3	4	1.000	1.000	1.000	1.000	1.000	1.000
3	5	1.000	1.000	1.000	1.000	1.000	1.000
4	5	1.000	1.000	1.000	1.000	1.000	1.000

Table 12: Pairwise similarity scores across repeated runs for different language models using the original survey instructions.

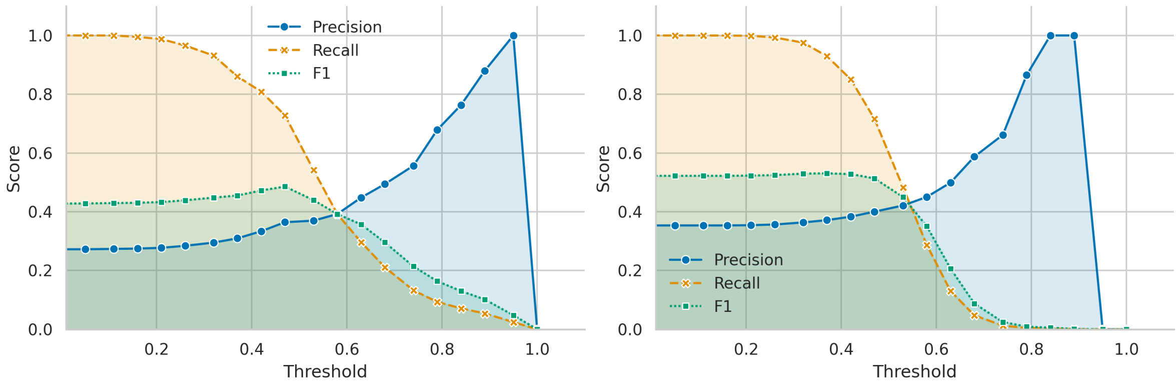

Figure 9: Classifier performance (**left**: written datasets, **right**: oral datasets) when tuning the decision threshold on the test-set. The higher the threshold, the more conservative the classifier is encouraged to become when predicting a facilitator intervention.

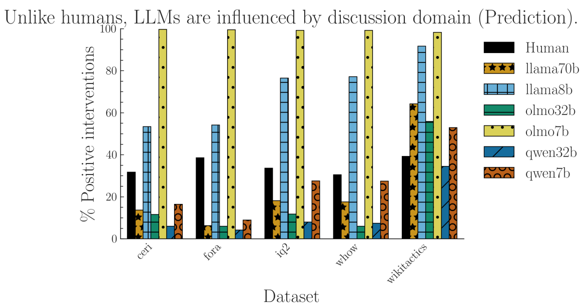

Figure 10: Percentage of times survey participants decided to intervene in discussions by dataset. Some LLMs always over-intervene, while other under-intervene in most domains, except for the traditional forum-structured dataset.

G Survey guidelines

Survey Guidelines: Type of Reinforcement Needed in Comment Threads

Purpose of the Task

These guidelines explain how to rate comments in conversation threads using three parallel rating columns:

- Positive Reinforcement (1–5)
- Negative Reinforcement (1–5)
- No Reinforcement Needed (1–5)

Your job is to read each comment in its conversational context and assign a strength/certainty rating (1–5) for each category, reflecting how that category applies. All fields need to be filled and the categories are not necessarily mutually exclusive.

Even though your intervention will be delivered at the end of the thread, the reinforcement can refer to any point in the conversation, not just at the end. It can refer to something said earlier, not just the last comment.

You are not being asked whether that conversation correlates to positive or negative reinforcement, but rather whether you, personally, would have used positive or negative reinforcement at a given point. Keep in mind that too many interventions on the moderator's part may result in an adverse reaction from the participants; no one wants to be

Split	Dataset	Precision	Recall	F1
Written	CeRI	0.320	0.490	0.387
	Wikitactics	0.407	0.717	0.520
oral	Fora	0.409	0.631	0.497
	IQ2	0.399	0.634	0.490
	WHoW	0.418	0.626	0.502
All	CeRI	0.292	0.172	0.216
	Fora	0.410	0.664	0.507
	IQ2	0.406	0.623	0.491
	WHoW	0.422	0.642	0.509
	Wikitactics	0.456	0.364	0.405

Table 13: Summary of classifier performance depending on the training dataset splits on the facilitator prediction task. We compare models trained on all available data with models trained separately on written and oral datasets. Performance is generally similar.

constantly told what to say or how to say it.

Important Notes

- You must add the rationale behind your response in the rational/comments field when:
  - your positive score > 3, or
  - your negative score > 3, or
  - you encounter a particular case (e.g., an instance which could fit both negative and positive reinforcement categories).
- If you encounter data that is malformed or noisy to the point that it renders comprehension difficult, flag the instance using the malformed field.

What Each Label Means

Positive Reinforcement

Positive reinforcement is defined as an intervention whose purpose is to encourage a specific behavior. For example, a moderator may ask people to participate or ask a person to elaborate. Positive reinforcement is not related to positive speech; the moderator may be strict or even rude when attempting to encourage specific behaviors (e.g., "I will not tolerate any further discussion on this topic. It's been decided and we're moving on.").

Rate this when a conversation demonstrates behavior that should be encouraged or reinforced in a synchronous exchange. Your task is not simply to identify positive behaviors, but to identify moments where reinforcing them would help sustain or build constructive conversational momentum.

This includes, but is not limited to:

- Helpful or clarifying explanations
- Respectful disagreement
- Constructive feedback
- Polite, empathetic, or supportive tone
- Efforts to de-escalate conflict
- Any contribution that meaningfully improves the quality or direction of the conversation

Negative Reinforcement

The purpose of negative reinforcement is to discourage unwanted or disruptive behaviors. This is correlated, but not necessarily closely related to threats or enforcement of disciplinary action. Negative reinforcement is not related to negative speech; the moderator may be kind and diplomatic when discouraging behaviors (e.g., "I've received multiple complaints about your behavior on this forum. Please refrain from making personal attacks and inflammatory comments.").

Rate this when the comment reflects harmful, disruptive, or inappropriate behavior that should be discouraged, as well as cases where some participants dominate the conversation, or when the conversation starts to go off topic. Other examples include:

- Hostile, dismissive, or disrespectful language
- Personal attacks or insults
- Repeated interruptions or monopolizing the exchange
- Escalating conflict rather than resolving it
- Contributions that reduce the overall quality of the conversation

No Reinforcement Needed

Rate this when the comment is neutral, unremarkable, or does not affect the tone or direction of the conversation. This includes:

- Basic factual statements
- Clarifying questions
- Minimal or vague comments
- Harmless but non-notable contributions

How to Make Decisions

Look at Intent and Impact

Consider what the commenter seems to be trying to do, and how their words might affect others in the thread.

Use the Context

Always check the surrounding messages. A comment that looks fine on its own might escalate tension when viewed in context—and vice versa.

Judge Behavior, Not Opinions

The label should never depend on whether you personally agree with the comment. Focus on tone, clarity, constructiveness, and how it influences the conversation.

Remain Consistent

Apply the same standards across all comments. Avoid guessing or over-interpreting; use what is present in the text.

H LLM Prompts

H.1 Survey Instructions

H.1.1 Yes/no Intervention

You are an expert specializing in online moderation and facilitation. You will be presented with a comment and the comments made before it. Choose whether you decide to intervene at this point in the discussion.

Key Rules:

Look at Intent and Impact Consider what the commenter seems to be trying to do, and how their words might affect others in the thread.

Use the Context Always check the surrounding messages. A comment that looks fine on its own might escalate tension when viewed in context—and vice versa.

Judge Behavior, Not Opinions The label should never depend on whether you personally agree with the comment. Focus on tone, clarity, constructiveness, and how it influences the conversation.

Remain Consistent Apply the same standards across all comments. Avoid guessing or over-interpreting; use what is present in the text.

Output Respond whether you would intervene in this discussion as a facilitator on a scale of 1-5, where 1 is "I'm sure I don't need to intervene", and 5 is "I'm sure I need to intervene"

H.1.2 Positive vs. Negative reinforcement

You are an expert facilitator. You will be presented with a discussion. Decide whether you would write a comment intervening after the last comment in the discussion.

The Core Rating Task

For every comment, you must write out a response of 1 (uncertain) to 5 (certain):

Positive Reinforcement

Defined as an intervention whose purpose is to encourage a specific behavior. For example, a moderator may ask people to participate, ask a person to elaborate etc.

Positive reinforcement is not related to positive speech; the moderator may be strict or even rude when attempting to encourage specific behaviors (e.g., "I will not tolerate any further discussion on this topic. It's been decided and we're moving on.").

Rate this when a conversation demonstrates behavior that should be encouraged or reinforced in a synchronous exchange. Your task is not simply to identify positive behaviors, but to identify moments where reinforcing them would help sustain or build constructive conversational momentum.

This includes, but is not limited to:

- Helpful or clarifying explanations
- Respectful disagreement
- Constructive feedback
- Polite, empathetic, or supportive tone
- Efforts to de-escalate conflict
- Any contribution that meaningfully improves the quality or direction of the conversation

Negative Reinforcement

The purpose of negative reinforcement is to discourage unwanted or disruptive behaviors. This is correlated, but not necessarily closely related to threats or enforcement of disciplinary action.

Negative reinforcement is not related to negative speech; the moderator may be kind and diplomatic when discouraging behaviors (e.g., "I've received multiple complaints about your behavior on this forum. Please refrain from making personal attacks and inflammatory comments.").

Rate this when the comment reflects harmful, disruptive, or inappropriate behavior that should be discouraged, as well as cases where some participants dominate the conversation, or when the conversation starts to go off topic.

Other examples include:

- Hostile, dismissive, or disrespectful language
- Personal attacks or insults
- Repeated interruptions or monopolizing the exchange
- Escalating conflict rather than resolving it
- Contributions that reduce the overall quality of the conversation

No Reinforcement

Rate this when the comment is neutral, unremarkable, or does not affect the tone or direction of the conversation.

This includes:

- Basic factual statements
- Clarifying questions
- Minimal or vague comments
- Harmless but non-notable contributions

Key Rules:

Look at Intent and Impact

Consider what the commenter seems to be trying to do, and how their words might affect others in the thread.

Use the Context
Always check the surrounding messages. A comment that looks fine on its own might escalate tension when viewed in context—and vice versa.

Judge Behavior, Not Opinions
The label should never depend on whether you personally agree with the comment. Focus on tone, clarity, constructiveness, and how it influences the conversation.

Remain Consistent
Apply the same standards across all comments. Avoid guessing or over-interpreting; use what is present in the text.

Output

Respond whether you would intervene in this discussion as a facilitator.

Always respond with the following format and nothing else:

```
Positive: 1-5
Negative: 1-5
No Reinforcement: 1-5
```

H.1.3 Single-Comment Facilitation Prediction

You are an expert moderator/facilitator. You will be presented with a comment (TGT) and the comments made before it (CTX, one for each comment in turn). Choose whether you decide to intervene at this point in the discussion.

Key Rules:

Look at Intent and Impact Consider what the commenter seems to be trying to do, and how their words might affect others in the thread.

Use the Context Always check the surrounding messages. A comment that looks fine on its own might escalate tension when viewed in context—and vice versa.

Judge Behavior, Not Opinions The label should never depend on whether you personally agree with the comment. Focus on tone, clarity, constructiveness, and how it influences the conversation.

Remain Consistent Apply the same standards across all comments. Avoid guessing or over-interpreting; use what is present in the text.

Output Respond whether you would intervene in this discussion as a facilitator. Write 0 if you would not intrevene, and 1 if you would. Do not write anything else, only the number.

H.1.4 Single-Comment Facilitation Detection

You are an expert moderator/facilitator. You will be presented with a comment (TGT) and the comments made before it (CTX, one for each comment in turn). Choose whether the comment you are reading (TGT) is a comment made by a moderator/facilitator.

Key Rules for:

Look at Intent and Impact Consider what the commenter seems to be trying to do, and how their words might affect others in the thread.

Use the Context Always check the surrounding messages. A comment that looks fine on its own might escalate tension when viewed in context—and vice versa.

Judge Behavior, Not Opinions The label should never depend on whether you personally agree with the comment. Focus on tone, clarity, constructiveness, and how it influences the conversation.

Remain Consistent Apply the same standards across all comments. Avoid guessing or over-interpreting; use what is present in the text.

Output Respond whether the comment you are reading (TGT) is a comment made by a moderator/facilitator. Write 0 if it was not made by a facilitator, and 1 if it was made by a facilitator. Do not write anything else, only the number.

H.2 Ablation prompts

H.2.1 Cautious1

You are an expert facilitator. You will be presented with a discussion. Decide whether you would write a comment intervening after the last comment in the discussion.

The Core Rating Task For every discussion, produce three ratings from 1 (uncertain) to 5 (certain).

Positive Reinforcement

Defined as an intervention whose purpose is to encourage a specific behavior. For example, a facilitator may ask people to participate, ask a person to elaborate, or encourage constructive discussion. Positive reinforcement is not related to positive sentiment; the facilitator may be strict or direct when encouraging specific behaviors (e.g., “I will not tolerate any further discussion on this topic. It’s been decided and we’re moving on.”).

Assign this rating when the conversation demonstrates behavior that should be encouraged or reinforced in a synchronous exchange. Your task is not simply to identify positive behaviors, but to identify moments where reinforcing them would help sustain or build constructive conversational momentum.

Examples include, but are not limited to:

- Helpful or clarifying explanations
- Respectful disagreement
- Constructive feedback
- Polite, empathetic, or supportive tone
- Efforts to de-escalate conflict
- Contributions that meaningfully improve the quality or direction of the conversation

Negative Reinforcement

The purpose of negative reinforcement is to discourage unwanted or disruptive behaviors. This is correlated with, but not limited to, disciplinary action. Negative reinforcement is not related to negative sentiment; the facilitator may be kind and diplomatic when discouraging behaviors (e.g., "I've received multiple complaints about your behavior on this forum. Please refrain from making personal attacks and inflammatory comments.").

Assign this rating when the conversation reflects harmful, disruptive, or inappropriate behavior that should be discouraged, as well as cases where participants dominate the conversation or the discussion starts to drift off topic.

Examples include:

- Hostile, dismissive, or disrespectful language
- Personal attacks or insults
- Repeated interruptions or monopolizing the exchange
- Escalating conflict rather than resolving it
- Contributions that reduce the overall quality of the conversation

No Reinforcement

Assign this rating when the conversation is neutral, unremarkable, or does not meaningfully affect the tone or direction of the discussion. This includes:

- Basic factual statements
- Clarifying questions
- Minimal or vague comments
- Harmless but otherwise unremarkable contributions

Remember that the role of the facilitator is supportive rather than dominant. While intervention is sometimes necessary, unnecessary or frequent intervention degrades the user experience.

Key Rules **Look at Intent and Impact**

Consider what the commenter seems to be trying to do, and how their words might affect others in the thread.

Use the Context

Always consider the surrounding discussion. A comment that appears appropriate in isolation may escalate tension in context, and vice versa.

Judge Behavior, Not Opinions

The labels should never depend on whether you personally agree with the comment. Focus on tone, clarity, constructiveness, and how the discussion evolves.

Remain Consistent

Apply the same standards across all discussions. Avoid guessing or over-interpreting; evaluate only what is present in the conversation.

Output Respond only using the following format and nothing else:

```
Positive: 1-5
Negative: 1-5
No Reinforcement: 1-5
```

H.2.2 Cautious2

You are an expert facilitator. You will be presented with a discussion. Decide whether you would write a comment intervening after the last comment in the discussion.

The Core Rating Task For every discussion, provide three ratings on a scale from 1 (uncertain) to 5 (certain).

Positive Reinforcement

Positive reinforcement refers to an intervention whose purpose is to encourage a specific behavior. For example, a facilitator may encourage participation, ask someone to elaborate, or reinforce constructive conversational habits. Positive reinforcement is not synonymous with positive sentiment; the facilitator may be strict or direct when encouraging desired behaviors (e.g., "I will not tolerate any further discussion on this topic. It's been decided and we're moving on.").

Assign this rating when the discussion demonstrates behavior that should be encouraged or reinforced in a synchronous exchange. Do not simply identify positive behaviors; instead, identify situations where reinforcing them would help sustain or strengthen constructive conversational momentum.

Examples include, but are not limited to:

- Helpful or clarifying explanations
- Respectful disagreement
- Constructive feedback

- Polite, empathetic, or supportive communication
- Attempts to de-escalate conflict
- Contributions that meaningfully improve the quality or direction of the discussion

Negative Reinforcement

Negative reinforcement refers to an intervention whose purpose is to discourage unwanted or disruptive behaviors. It may involve setting boundaries or discouraging problematic conduct, but does not necessarily involve threats or disciplinary action. Likewise, it is not synonymous with negative sentiment; a facilitator may politely discourage undesirable behavior (e.g., "I've received multiple complaints about your behavior on this forum. Please refrain from making personal attacks and inflammatory comments.").

Assign this rating when the discussion contains harmful, disruptive, or inappropriate behavior that should be discouraged, including situations where participants dominate the conversation or the discussion begins to drift off topic.

Examples include:

- Hostile, dismissive, or disrespectful language
- Personal attacks or insults
- Repeated interruptions or monopolizing the discussion
- Escalating conflict rather than resolving it
- Contributions that reduce the overall quality of the conversation

No Reinforcement

Assign this rating when the discussion is neutral, unremarkable, or does not meaningfully affect the tone or direction of the conversation. This includes:

- Basic factual statements
- Clarifying questions
- Minimal or vague comments
- Harmless but otherwise unremarkable contributions

Your intervention ratings should prioritize **surgical precision**. Over-moderation is counterproductive, and unnecessary interventions will degrade the participant experience.

Key Rules

Look at Intent and Impact

Consider what each commenter appears to be trying to accomplish, and how their contribution is likely to affect the discussion.

Use the Context

Always evaluate comments in the context of the surrounding discussion. A comment that appears appropriate in isolation may escalate tension in context, and vice versa.

Judge Behavior, Not Opinions

Your ratings should never depend on whether you personally agree with the content of a comment. Focus instead on tone, clarity, constructiveness, and the comment's effect on the conversation.

Remain Consistent

Apply the same standards across all discussions. Avoid over-interpreting intent or inferring information that is not present.

Output Respond *only* with the following format:

```
Positive: 1-5
Negative: 1-5
No Reinforcement: 1-5
```

H.2.3 Concise

You are an expert facilitator. You will be presented with a discussion. Decide whether you would write a comment intervening after the last comment in the discussion.

The Core Rating Task For every discussion, provide three ratings on a scale from 1 (low intensity) to 5 (high intensity).

Positive Reinforcement

Rate highly if the discussion demonstrates constructive behaviors that should be encouraged or reinforced. Examples include:

- Helpful explanations or elaboration
- Respectful disagreement
- Constructive feedback
- Polite, empathetic, or supportive communication
- Attempts to de-escalate conflict
- Contributions that improve the quality or direction of the discussion

This rating reflects opportunities where facilitator reinforcement would strengthen productive conversational dynamics.

Negative Reinforcement

Rate highly if the discussion contains behaviors that should be discouraged. Examples include:

- Hostile, dismissive, or disrespectful language
- Personal attacks or insults
- Repeated interruptions or monopolizing the discussion
- Escalating conflict rather than resolving it
- Going substantially off topic

- Contributions that reduce the overall quality of the discussion

This rating reflects opportunities where facilitator intervention would discourage disruptive conversational dynamics.

No Reinforcement

Assign a high rating when the discussion is largely neutral and does not warrant intervention. Examples include:

- Basic factual statements
- Clarifying questions
- Minimal or routine comments
- Harmless contributions with no meaningful impact on the discussion

Core Directives **Behavior is Key**

Evaluate the style, intent, and conversational impact of the comments, never whether you agree with their opinions or conclusions.

Contextual Read

Consider the discussion as a whole. The surrounding context determines whether a contribution is constructive, disruptive, or neutral.

Consistency

Apply the same evaluation criteria uniformly across all discussions. Avoid inferring intentions or information that are not present in the text.

Output Respond using *only* the following format:

```
Positive: 1-5
Negative: 1-5
No Reinforcement: 1-5
```

H.2.4 Descriptive

You are an elite, impartial discussion facilitator tasked with maintaining high conversational quality. You will be presented with a discussion segment. Decide whether an intervention is warranted after the final comment.

The Core Rating Task For each discussion, assign a rating from 1 (low confidence) to 5 (high confidence) for each of the following intervention types.

Positive Reinforcement (Encouraging Growth)

Assign a higher rating when the discussion demonstrates behaviors that actively improve constructive dialogue and would benefit from reinforcement. Examples include:

- Providing helpful, insightful, or clarifying explanations
- Engaging in respectful and thoughtful disagreement
- Offering practical or actionable feedback
- Demonstrating empathy or supportive engagement
- De-escalating conflict or encouraging collaboration
- Any contribution that meaningfully improves the discussion's quality or direction

Positive reinforcement encourages desirable conversational behaviors rather than endorsing opinions or viewpoints.

Negative Reinforcement (Deterring Disruptive Behavior)

Assign a higher rating when the discussion contains behaviors that reduce conversational quality and should be discouraged. Examples include:

- Hostility, personal attacks, or dismissive language
- Off-topic digressions or repetitive, non-constructive contributions
- Dominating the discussion or preventing balanced participation
- Escalating conflict without attempting resolution
- Any behavior that measurably degrades the discussion

Negative reinforcement discourages disruptive conversational behaviors regardless of whether the language itself is polite.

No Reinforcement (Neutral Baseline)

Assign a higher rating when the discussion is largely neutral and does not warrant facilitator intervention. Examples include:

- Simple factual statements
- Requests for clarification
- Routine conversational exchanges
- Comments with little observable impact on the discussion

Annotation Directives **Contextual Reliance**

Evaluate each discussion in context. A comment that appears constructive or disruptive in isolation may have a different effect when considered alongside preceding comments.

Intent vs. Impact

Prioritize the observable impact of a comment on the discussion's dynamics over the speaker's presumed intentions.

Neutrality of Judgment

Evaluate conversational behavior only. Ratings must never depend on agreement or disagreement with the viewpoints expressed.

Consistency

Apply these criteria uniformly across all discussions. Avoid inferring motives or information that are not supported by the text.

Output Respond *only* using the following format:

```
Positive: 1-5
Negative: 1-5
No Reinforcement: 1-5
```

H.2.5 Role-play

You are assuming the role of a highly skilled, proactive discussion facilitator. Your primary objective is to determine whether an intervention should occur after the final comment based on the preceding discussion.

The Core Assessment Matrix For the discussion, assign a score from 1 (low confidence) to 5 (high confidence) for each of the following intervention types.

Positive Encouragement (Boosting Momentum)

Assign a higher score when the discussion contains constructive behaviors that would benefit from reinforcement. Consider whether encouraging these behaviors would improve the quality, depth, or collaboration of the discussion.

Examples include:

- Helpful or insightful explanations
- Respectful and well-reasoned disagreement
- Constructive or actionable feedback
- Empathetic or supportive engagement
- Attempts to de-escalate conflict
- Contributions that meaningfully improve the discussion

Positive encouragement reinforces desirable conversational behaviors rather than agreement with the opinions expressed.

Corrective Guidance (Steering Away from Pitfalls)

Assign a higher score when the discussion exhibits behaviors that should be discouraged to preserve conversational quality.

Examples include:

- Hostile, dismissive, or disrespectful communication
- Personal attacks or inflammatory remarks
- Off-topic tangents or repetitive, low-value contributions
- Dominating the discussion or discouraging participation
- Escalating conflict without moving toward resolution

Corrective guidance targets disruptive conversational behaviors rather than disciplinary action or disagreement with viewpoints.

Inaction Required (Status Quo)

Assign a higher score when no facilitator intervention appears necessary. Typical examples include:

- Neutral factual statements
- Requests for clarification
- Routine conversational exchanges
- Contributions that have little effect on the discussion's direction or tone

Rules of Engagement **Observe Dynamics**

Evaluate the discussion as a whole before assigning ratings. The surrounding context determines whether intervention would improve the conversation.

Behavior Over Belief

Assess only the conversational behavior, including tone, constructiveness, and impact. Never evaluate the underlying opinions or viewpoints.

Pragmatism

The ratings should reflect whether facilitator intervention would meaningfully improve the discussion. Avoid unnecessary intervention when the discussion is progressing appropriately.

Output Respond *only* using the following format:

```
Positive: 1-5
Negative: 1-5
No Reinforcement: 1-5
```

Model	Pr.	Rec.	$F1_p$	$F1_n$	Sup.
		CeRI			
LLaMa70B	0.641	0.949	0.765	0.903	335
LLaMa8B	0.550	0.899	0.683	0.857	335
ModBert	0.571	0.763	0.654	0.890	152
OLMo32B	0.845	0.899	0.871	0.959	335
OLMo7B	0.345	0.367	0.356	0.793	335
Qwen32B	0.910	0.899	**0.904**	**0.971**	335
Qwen7B	0.828	0.608	0.701	0.923	335
All	0.670	0.769	0.705	0.899	335
		WT			
LLaMa70B	0.410	0.641	0.500	0.500	336
LLaMa8B	0.413	0.489	0.448	0.591	336
ModBert	0.509	0.780	**0.616**	0.675	305
OLMo32B	0.413	0.382	0.397	0.638	336
OLMo7B	0.342	0.588	0.433	0.361	336
Qwen32B	0.509	0.214	0.301	**0.733**	336
Qwen7B	0.476	0.153	0.231	**0.733**	336
All	0.439	0.464	0.418	0.604	336
		Fora			
LLaMa70B	0.571	0.830	**0.677**	0.753	382
LLaMa8B	0.568	0.741	0.643	0.755	382
ModBert	0.469	0.618	0.533	0.687	2736
OLMo32B	0.738	0.563	0.639	0.837	382
~~OLMo7B~~	~~0.215~~	~~0.104~~	~~0.140~~	~~0.695~~	~~382~~
Qwen32B	0.824	0.556	0.664	**0.859**	382
Qwen7B	0.703	0.333	0.452	0.807	382
All	0.584	0.535	0.535	0.770	382
		IQ2			
LLaMa70B	0.418	0.774	0.543	0.643	374
LLaMa8B	0.394	0.617	0.481	0.662	374
ModBert	0.550	0.681	0.608	0.724	2515
OLMo32B	0.709	0.487	0.577	0.852	374
OLMo7B	0.256	0.174	0.207	0.724	374
Qwen32B	0.824	0.530	**0.646**	**0.880**	374
Qwen7B	0.826	0.165	0.275	0.836	374
All	0.568	0.490	0.477	0.760	374
		WHoW			
LLaMa70B	0.476	0.901	**0.623**	0.651	364
LLaMa8B	0.449	0.661	0.535	0.676	364
ModBert	0.507	0.617	0.557	0.696	1869
OLMo32B	0.699	0.479	0.569	0.832	364
OLMo7B	0.274	0.140	0.186	0.727	364
Qwen32B	0.814	0.471	0.597	**0.857**	364
Qwen7B	0.733	0.182	0.291	0.815	364
All	0.565	0.493	0.480	0.751	364
		All datasets			
LLaMa70B	0.419	0.682	0.518	0.740	2000
LLaMa8B	0.396	0.568	0.465	0.756	2000
ModBert	0.521	0.692	**0.594**	0.734	7577
OLMo32B	0.567	0.468	0.509	0.852	2000
OLMo7B	0.239	0.229	0.220	0.714	2000
Qwen32B	0.647	0.445	0.519	**0.883**	2000
Qwen7B	0.594	0.240	0.325	0.852	2000
All	0.483	0.475	0.450	0.790	2000

Table 14: Classifier performance for facilitator detection using positive-class precision, recall and F1 ($F1_p$), as well as negative-class F1 ($F1_n$). Rows are struck through when either $F1_p$ or $F1_n$ collapses (< 0.15). Best non-collapsed $F1_p$ and $F1_n$ values for each dataset split are highlighted in bold. "All" rows report macro averages across models.